Individual differences in the computational processes that support face recognition

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Face recognition ability varies substantially across the population, yet the cognitive and computational processes that give rise to this variation remain poorly understood. Here, we combined approaches from experimental, differential and mathematical psychology to test how individual differences in face recognition ability relate to latent parameters of the decision-making processes involved in face recognition. Participants (N = 300) completed a face memory task with learned and novel faces, together with independent measures of face recognition ability (Cambridge Face Memory Test) and general object recognition ability (car recognition). We modelled task performance with an evidence-accumulation model that translates accuracy and response times into latent parameters, including drift rate (the signal-to-noise ratio of the evidence) and threshold (response caution). Higher face recognition ability was associated with greater drift rate discrimination between correct and incorrect response accumulators in the face memory task, for both learned and novel faces and to a similar degree, consistent with expertise effects in other domains. This association remained largely unchanged when accounting for car recognition ability, suggesting that it is at least partly tied to face processing itself. Higher face recognition ability was also associated with greater response caution, which may reflect better calibration to task demands. Although the associations were modest in size, these findings suggest that individual differences in face recognition ability are reflected in distinct computational mechanisms, namely more efficient extraction of information from faces and a more cautious decision policy. This approach enables a more explicit and computational basis for cognitive theories of face recognition to be developed, as well as for testing whether these mechanisms operate similarly across the full spectrum of face recognition abilities.

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1. Introduction

Being able to quickly and accurately recognise other people is a central aspect of everyday social interactions, as well as more specific tasks, such as eye-witness identification and passport control checks. However, face recognition ability varies across the population. For instance, people with developmental prosopagnosia show relatively poor face recognition ability (Dobel et al., 2007; McConachie, 1976), whereas so-called "super-recognisers" show a remarkably high face recognition ability compared to the average person (Bobak et al., 2016; Russell et al., 2009). Even if we ignore these two extreme cases, face recognition ability varies throughout the general population, especially when identifying unfamiliar individuals (Dowsett & Burton, 2015; White et al., 2014). Unsurprisingly, therefore, research has attempted to understand the mental processes that determine variation in face recognition ability as one way to probe the structure of social perception and cognition (Rhodes et al., 2014; Wang et al., 2012; Yovel et al., 2014). Here, in this project, we move this research field forward by combining approaches from experimental, differential and mathematical psychology, to investigate how individual differences in face recognition ability relate to the computational processes that support face recognition.

Attempts to understand differences between individuals has a long history in psychological research. Indeed, differential psychology represents a major branch of psychological research that aims to understand individual differences in thought, behaviour and ability (Lubinski, 2000; Marsh & Boag, 2013). In the domain of face recognition, there is a growing interest in investigating individual differences across the entire spectrum of face recognition ability, rather than focusing on specific subgroups, such as "super-recognisers" (White & Kitchen, 2022). To this end, several assessment tools have been designed to measure different facets of face recognition performance (Burton et al., 2010; Duchaine & Nakayama, 2006; Russell et al., 2009). For example, the Cambridge Face Memory Test [CFMT; Duchaine & Nakayama (2006)] measures how many previously seen faces individuals can accurately recognise, with performance varying across the general population and being poorer in individuals with developmental prosopagnosia. Furthermore, the CFMT shows high internal consistency (Bowles et al., 2009; Wilmer et al., 2010), measures a single underlying ability across a wide range of performance levels (Cho et al., 2015), and yields reasonably stable scores when the same individuals are retested months or years later (Murray & Bate, 2020; Wilmer et al., 2010). This suggests it captures a stable ability that is relatively specific to faces. As such, these tools appear to have sufficient validity to represent a marker of an individual's face recognition ability.

Using these tools as a marker of face recognition ability, past research has estimated the association between the performance on these tests and broader perceptual and cognitive processes (McCaffery et al., 2018; Noyes et al., 2018). One line of research has studied the extent to which individual differences in face processing reflect domain-general or more domain-specific variation in perceptual and cognitive abilities (McCaffery et al., 2018; Verhallen et al., 2017). These findings show that face and general object recognition ability are positively, but modestly associated (Dennett et al., 2012a; Richler et al., 2019a; Wilmer et al., 2012). Nonetheless, they remain constantly weaker than associations between different face recognition tasks (Fysh et al., 2020; McCaffery et al., 2018; Wilmer et al., 2010). One interpretation of these results is that face recognition tasks rely partly on a specialised, domain-specific set of mental processes (Richler et al., 2019b), in addition to more domain-general perceptual abilities.

The use of directly observed or manifest measures, such as reaction time and accuracy, has made a valuable contribution to understanding the relationship between psychological processes in general and in the context of face perception. However, performance on such tests remains a distant proxy of the underlying cognitive processes at work during face recognition, which makes them far from ideal for capturing individual differences in cognitive processes (Haines et al., 2020). An alternative approach to this problem has emerged in the domain of mathematical psychology and involves using a computational model to estimate latent computational parameters, which can be closer to the cognitive processes of interest (Yarkoni, 2022). For instance, evidence-accumulation models translate the distribution of response times for correct and incorrect responses into parameters reflecting latent computational processes that are supposed to underlie the decision-making process (see Ratcliff et al. (2016) for a review).

One example of such evidence-accumulation models is the linear ballistic accumulator (LBA) model (Figure 1) (Brown & Heathcote (2008)). In two-alternatives forced choice tasks, this model describes the accumulation of evidence (drift rate parameter) towards the two alternative options. Once the accumulation of evidence crosses a threshold, the corresponding response is then chosen. Importantly, parameters from this model can be interpreted in psychologically meaningful ways: the drift rate parameter reflects the signal-to-noise ratio provided by the stimulus and the threshold parameter reflects the response caution towards a response (the amount of evidence needed to trigger a particular response; Myers et al. (2022)). As a consequence, one can then study how variation between people in these latent computational parameters, rather than manifest or observed measures, relates to key aspects of social behaviour, such as face recognition ability (White et al., 2016).

Approaches from differential psychology have historically run alongside, and largely separate to, research traditions from experimental psychology, which aim to estimate the average effect (aggregated across individuals) of an experimental manipulation (Myers & Hansen, 2006). For example, we previously used an experimental approach in combination with evidence accumulation modelling to study how familiarity impacts the computational processes that support face recognition (Fournier et al., 2026). Different types of familiarity (visually familiar faces vs. famous faces) had a similar impact on the computational processes involved in the decision-making process of face recognition. We found that evidence accumulated more rapidly (drift rate) for novel faces. Further, response caution (threshold) was stronger for novel than learned faces. Although we showed that these results could be replicated in both laboratory and online settings, the work aimed to estimate an average effect across a relatively small number of individuals (N~30), rather than investigate individual differences. As such, Fournier et al. (2026) were unable to estimate how variation in face recognition ability between individuals shapes the computational processes that determine how face recognition judgements are made.

In the present study, therefore, we combined approaches from differential, experimental and mathematical psychology to investigate how latent computational parameters in face recognition relate to face recognition ability. To do so, we used our prior face memory task (Fournier et al., 2026) to compute key latent parameters, whilst also measuring face recognition ability, using the

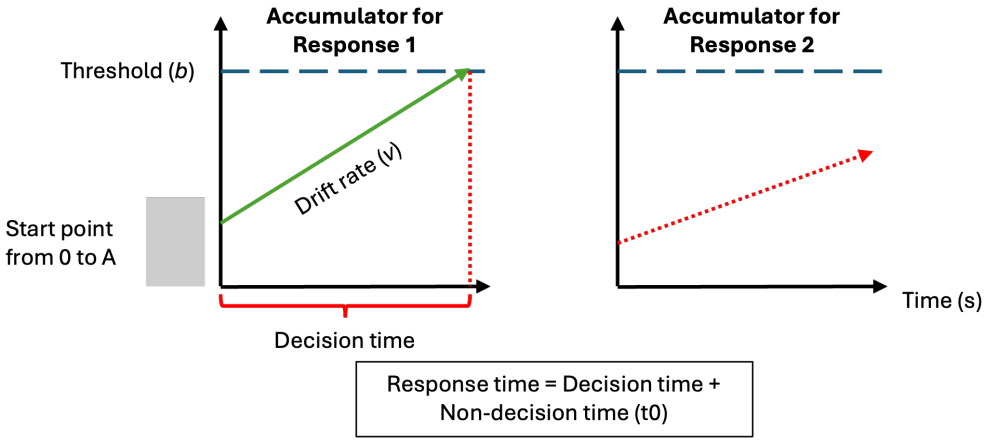


Figure 1. Schematic of the linear ballistic accumulator (LBA) model

This diagram represents the accumulation process one trial of a perceptual decision task for the linear ballistic accumulator (LBA; Brown & Heathcote (2008)). Participants are required to chose between two options (Response 1 and Response 2). In a LBA model, there is a separated accumulation of evidence (v) for each response option. In this example, the correct response is Response 1. The accumulator on the left is the accumulator for the correct response (Response 1), and the green line in represents the rate at which the evidence accumulates in favor of that response option. The accumulator on the right is the accumulator for the incorrect response (Response 2), and the dash red line represents the rate at which the evidence accumulates in favor of this response option. Once the accumulated evidence crosses the threshold (b) of a response option, the corresponding response is chosen (in this case Response 1).

CFMT, (Duchaine & Nakayama, 2006) and general object recognition ability, using the Cambridge Car Memory Test [CCMT; Dennett et al. (2012a)]. We also studied a considerably larger sample of participants (N~300) than our prior work to enable more robust estimates of between participant individual differences. Based on this research design, we were then able to estimate individual differences in drift rate and threshold as a function by face familiarity (i.e., novel vs visually familiar faces), as well as the extent to which such associations reflect the operation of domain-specific (face) or domain-general (object) processes. In general, and based on past research on the development of expertise (Dutilh et al., 2009; 2011; Parker et al., 2026; Reinhartz et al., 2023), we expect drift rate discrimination (the quality of the evidence accumulation) to positively correlate with face recognition ability, but it is less clear how threshold (response caution towards a response) may vary, as it is plausible that better face recognition could result in being more or less cautious.

2. Method and Materials

2.1. Pre-registration and open science

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study (Simons et al., 2017). The research question, hypotheses, experimental design, planned analysis, and exclusion criteria were preregistered (https://osf.io/b74gt/overview). Raw data and analysis code are available on Github (https://github.com/cbs-lab/face-familiarity-eam-iid). Models objects can be found on OSF (https://osf.io/ku9ce/overview).

2.2. Participants

All participants were recruited on the Prolific platform (https://www.prolific.com). Participants had to be located in the United Kingdom or in the United States, be aged between 18-65 years old, with normal or corrected-to-normal vision and speak fluent English. Furthermore, Prolific offers the option to chose the approval rate of the participants willing to take part in the study. The approval rate is the percentage for which participants has been approved in previous experiment. An approval rating between 95% and 100% was chosen in our study to try to ensure high data quality. 383 participants completed the experiment. Following our preregistered exclusion criterion, 2 participants were excluded because their percentage of missed trials was higher than 40%, 3 because their percentage of RT outside the established range (0.2 to 2 seconds) were higher than 25% and 78 were removed because the overall accuracy was lower than 55%. In the end, data from 300 participants (141 men and 159 women) were retained for the analysis and the LBA modeling. Ages ranged from 19 to 64 ($M_{age} = 42$, $SD_{age} = 11.2$). Following Prolific payment principles, each participant received 9£ for the completion of the study. This experiment was accepted by the ethics committee of the Federal Institute of technology (ETH) Zurich.

2.3. Material

Participant were asked to complete the study using a desktop computer. Beforehand, the paradigm was tested on three different operative systems (PC, Macintosh and Linux) and on three web browsers (Google Chrome, Safari and Firefox). Participant were aware that using any other combination of operative systems and web browsers might lead to the inability to complete the experiment and were asked to follow these directives to ensure a consistent procedure and participant experience.

2.4. Procedure

The experiment was divided into three experimental tasks. A face recognition task was used to estimate the LBA parameters from manifest measures (accuracy and response times). Face and car recognition ability were assessed using the Cambridge Face Memory Test and the Cambridge Car Memory Test. Participants always first underwent the face recognition task and the order of two memory tests was counterbalanced.

2.4.1. Face recognition task

The face recognition task was divided into two distinct parts: a familiarization part and a recognition part. During the familiarization part, unknown identities were presented on the screen

and participants were instructed to memorize the identity of the faces. One familiarization trial consisted of the presentation of 6 face images, one after the other, showing the same identity but varying in viewing points (3/4 left view, frontal view and 3/4 right view) and the colorimetry (colorful image, gray scale image). The duration of each image presentation was set to 0.75 second and each participant underwent 30 familiarization trials (for 30 unknown identities). During the familiarization, we introduced catch trials to ensure participants attention. After some familiarization trials, a gray scale image of a person's face was presented on the screen. Participants were asked to press the "q" key if the face belonged to the last seen identity or the "p" key if it was not. The faces seen during the familiarization were considered the "Learned faces" within-subject condition.

In the recognition part, participants completed three blocks of 90 recognition trials. Each block was separated from another by a short break. Half of the trials were learned identities trials and the other half were new, previously unseen identities trials. In each trial, a gray scale image of a face was shown on the screen. Participants were asked to choose if they had previously seen this face in the familiarization part or not. If they did, the should press the "q" key and if they did not, the "p" key. Participants were asked to respond as quickly and accurately as possible. It is important to note that participants never saw two pictures of the same learned identity in the same block and never saw the same picture (3/4 left view, frontal view, or 3/4 right view) of a learned identity during the entire recognition part. Furthermore, each novel identity was only presented once in the entire recognition part. Response time [s] and accuracy were collected after each recognition trial. A response time limit of 2 seconds has been set for each recognition trial. If participants took longer than 2 seconds to press a response key, the text "Too late!" appeared in red on the screen and the trial was discarded, but not replaced.

2.4.2. Cambridge Memory Tests

The method of the CFMT was described in Duchaine & Nakayama (2006). Participants learned 6 Caucasian male faces in three views and had to discriminate them later from distractor faces (no time response limit, three-alternative forced-choice task, Figure 3 a). Regarding the CCMT, the method has been described in Dennett et al. (2012a) and has a highly similar structure, format and scoring as the CFMT, but uses cars instead of faces as stimuli. The CCMT shows high internal consistency (Dennett et al., 2012a) and has been widely used alongside the CFMT as a measure of object recognition ability, for example in studies of developmental prosopagnosia (Gray et al., 2019; Murray & Bate, 2020) and of the genetic basis of face recognition (Shakeshaft & Plomin, 2015).

2.5. Stimuli

The face images used in the faces recognition task consisted of images selected from two different data sets of face images: the Face Research Lab London Set (DeBruine & Jones, 2017) and the Color FERET Database (https://www.nist.gov/itl/products-and-services/color-feret-database). Each of these data sets is composed of images of different identities faces taken from different angle points. 102 identities were selected from the Face Research Lab London Set and 51 from the Color FERET Database. Following this selection, we removed 27 identities (21 from the Face Research Lab London Set and 6 from the Color FERET Database) due to the presence of identifiable features (i.e. skin marks, make-up…). For each selected identity, three pictures of the face at three different view points (3/4 left view, frontal view and 3/4 right view) were cropped following the outline of the face. To ensure that the stimuli that we used are reproducible, we used the webmorphR R Package (DeBruine, 2022) to crop the face images. After the cropping, we removed 5 identities for which the cropping process resulted into "over-cropped" pictures (i.e. cropped nose or cropped eyebrows). In total, the data set of face images used in Experiment 1 contained 726 images: one set of three colour face images and one set of three gray scaled face images for each of the 121 identities. The code used to crop the original pictures can be found on OSF (https://osf.io/ku9ce/files/osfstorage).

2.6. Evidence accumulation modeling

The model and priors specification as well as the model estimation are all identical to the one described in Fournier et al. (2026). The priors specifications are described in Supplementary Materials.

2.6.1. Model specification

A hierarchical LBA models was fit to the accuracy and response times of the main task. The model specifications are outlined in our preregistration (Table 1). The LBA framework assumes a separate evidence accumulator for each response alternative, with potentially different parameter values. Accordingly, one accumulator corresponded to the "Learned" response and the other to the "Novel" response. Threshold was allowed to vary by participants' response (Learned/Novel) and mean drift rate varied as a function of both stimulus type (Learned/Novel) and weather the accumulator corresponded to the correct response for that stimulus (TRUE vs. FALSE). For example, when a Learned stimulus was presented, the Learned accumulator was designed as TRUE, whereas the Novel accumulator was designed as FALSE. The difference in drift rate between the TRUE and FALSE accumulators reflects the quality of information supporting a decision, with larger differences indicating stronger and more discriminative evidence accumulation (Lewandowsky & Oberauer, 2018). The standard deviation of the TRUE accumulator was allowed to vary according to accumulator match status, whereas the standard deviation of the FALSE accumulator was constrained to be constant to ensure model identifiability (Donkin et al., 2009). In the preregistered model, a single start-point parameter (A), and single non-decision time parameter (t0) were esti-

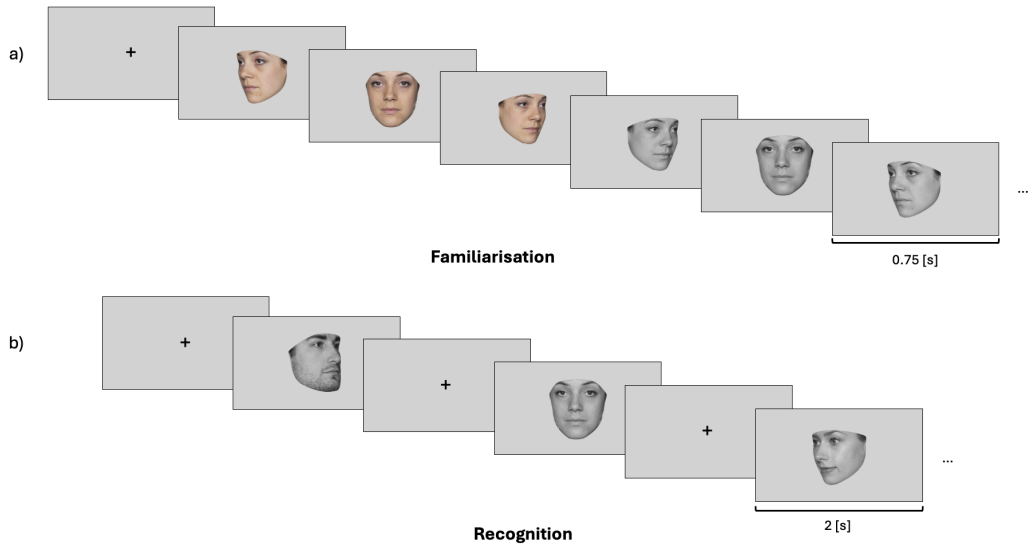


Figure 2. Experimental procedure of the face recognition task

The experimental procedure was divided into two parts: a familiarization part and a recognition part. a) In the familiarization part, participants were asked to memorize the face identity of 30 identities. For each identity, participants looked at 6 pictures depicting the same identity with different face orientation (3/4 left view, frontal view and 3/4 right view) and different colorimetry (color and gray scale images) in the order presented here: 3/4 left view, frontal view and 3/4 right view in color, followed by 3/4 right view, frontal view and 3/4 left view in gray scale. b) In each trial of the recognition part, participants had to decided if they recognize the face shown on the screen from the familiarization part. If they did, they had to press the "q" key and if the did not, the "p" key. They completed three blocks of 60 trials (half of them were from the familiarization part, half were novel faces). For the recognition part, only gray scale images of faces were used (for both learned faces and novel faces).

mated. Alternative model specifications that permitted variation in these parameters are reported in the Supplementary Materials.

Table 1. Formula specification for the pre-registered linear ballistic accumulator model.

Parameter	Label	Condition	Response	Accuracy	Fixed
Start point	A	no	no	no	-
Mean drift rate	v	yes	no	yes	-
SD drift rate	sd_v	no	no	yes	-
SD false drift rate	false.sd_v	-	-	-	1
Threshold	b	no	yes	no	-
Non-decision time	t0	no	no	no	-

Condition = training condition (Learned vs. Novel); Response = key-press response (same vs. different); Accuracy = (true/correct vs false/incorrect); SD = standard deviation.

2.6.2. Model estimation

Bayesian model estimated was carried using the EMC2 R package (Stevenson et al., 2026). Priors were chosen to be weakly-informative, meaning that they have little influence on the estimation (for more details regarding the priors of each model, see Supplementary Materials). Two sampling steps were used to estimate the parameters. First, a separate estimation of the fixed effects for each

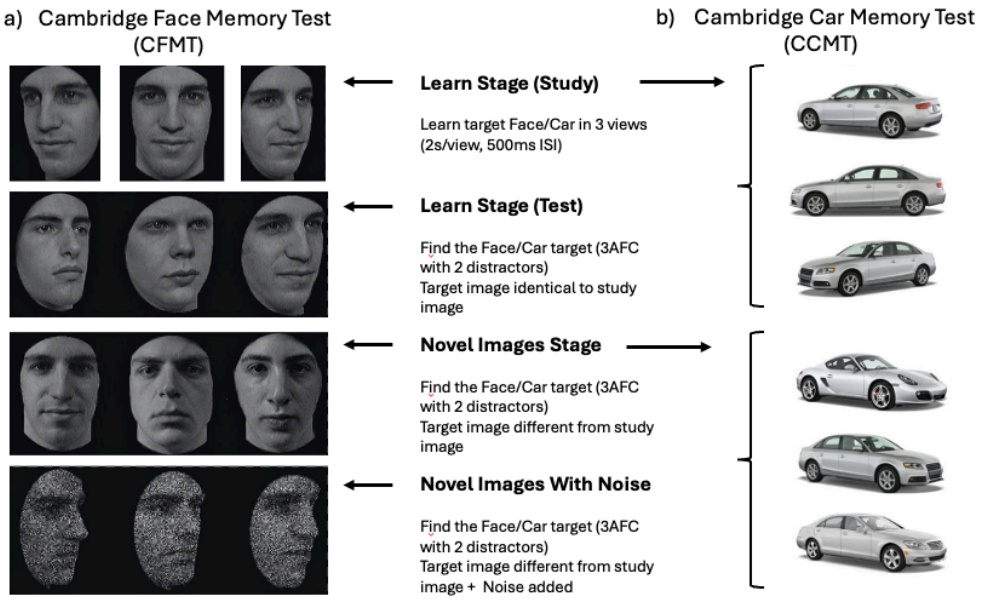


Figure 3. The two Cambridge Memory Tests

Illustration of (a) the Cambridge Face Memory Test (CFMT; ref Duchain & Nakayama, 2006) and (b) the Cambridge Car Memory Test (CCMT; red Dennett et al., 2012a). These two tests are highly similar, differing only in the category of the presented stimuli. Both tests are composed of three stages: Learn (including Study and Test), Novel Images and Novel Images with Noise. In the Learn stage, participants learn target faces/cars and are then tested on the recognition of the identical images to the learned targets. In the Novel Images and Novel Images with Noise stages, recognition targets are different pictures of the same learned targets, but differing in views, lightning or both. For the CCMT (b), the car pictures showed here are not the one used in the CCMT and served only as a representation of the stimuli used in the Learn stage and the Novel stages of the CCMT. This illustration has been taken and slightly modified from Dennett et al. (2012b). *3AFC = three-alternative forced choice; ISI = interstimulus interval.*

participant was conducted. Parameter estimations were then used as the starting point for the full hierarchical parameter estimation.

2.6.3. Model fit

To ensure that the fit of our model can capture most of the main trends in the data, graphical summaries of the posterior predictions were examined. Convergence plots and posterior predictive checks can be found in Supplementary Materials.

2.7. Data analysis

2.7.1. An overview of our general analytical strategy

We used two main analytical approaches to address our research questions. First, we used correlations and data visualizations to characterize the zero-order relationships between parameters of interest (drift rate and threshold) from the LBA model with face and object recognition ability. We used a "classical" correlation approach to generate point estimates of the correlation value and present those alongside scatter plots, as well as a Bayesian plausible value correlation approach to estimate the uncertainty in the estimated correlation values. Second, we used linear regression to estimate how LBA parameters vary as a function of face recognition ability, both before and after adjusting for general object (car) recognition ability. The details of the analysis plan were preregistered (https://osf.io/b74gt/overview) and more specific details are presented below.

2.7.2. LBA parameters

In order to make inference about the potential associations between LBA parameters and recognition abilities, subject specific estimates and group estimates posterior distributions were computed. Group estimates were contrasted across conditions (learned vs novel) to establish that the basic experimental effect found in Fournier et al. (2026) is replicable at the group level. To summarize the distribution of parameters of interest, we report the median of the posterior distribution together with 95% quantile interval of the posterior distribution. Assessment of individual differences in parameter estimates was pursued using subject specific estimates.

2.7.3. Classical correlational analysis

To assess the relationship between LBA parameters and Cambridge Memory tests, we computed the median value of each parameter's posterior distribution with the z-scored scores of each Memory Tests. For the threshold parameter, we computed the median of the two threshold parameters (threshold for learned and novel responses). For the drift rate parameter, we computed the median of the median of the six drift rate parameters. Drift rate (TRUE) reflects the accumulation of evidence when the response was correct (accumulation of signal) whereas drift rate (FALSE) represents the accumulation of evidence when the response was incorrect (accumulation of noise). Drift rate (TRUE-FALSE) is the subtraction drift rate (TRUE) - drift rate (FALSE) and represent the signal to noise ratio provided by the stimulus. For each stimulus condition (learned and novel), the median drift rate (TRUE), drift rate (FALSE) and drift rate (TRUE-FALSE) was computed. All the these median parameters were then correlated with the z-transformed scores of each Memory Test (Cambridge Face Mmeory Test and Cambridge Car Memory Test).

2.7.4. Bayesian plausible value correlational analysis

In addition to the classical correlational analysis described above, we conducted a Bayesian plausible value correlation approach (Ly et al., 2017; Matzke et al., 2017). Instead of using point estimates (median value of each parameter's posterior distribution), this approach permits the estimation of uncertainty in likely correlational values by correlating the parameter estimation from each sample (or "draw") from the posterior distribution with the memory scores. A distribution of the plausible correlations between each parameter and the memory scores can then be obtained. These distributions can then be compared between stimuli conditions (Learned vs Novel) or memory scores (face vs car) using difference contrasts. By summarising the posterior distribution of these correlation values, this approach naturally lends itself to estimates of the uncertainty of the estimated correlation values.

2.7.5. Bayesian linear regression

Using Bayesian linear regression allowed us ot investigate how individual differences in face recognition ability predicted the median drift rate values. Based on our pilot results, we preregistered

two Bayesian linear regression models to quantify the extent to which drift rate vary as a function of face recognition ability alone and when accounting for car recognition ability (see Table 2 for the details of each regression models). More specifically, the first model (Face model) predicts the median drift rate value as a function of the interaction effect between condition (learned vs novel) and accuracy (TRUE vs FALSE) and the main effect of face recognition ability (z-transformed scores to the Cambridge Face Memory score). An index coding approach (no grand intercept) was used, meaning that each sub-term of the interaction term has its own intercept (Learned_TRUE, Learned_FALSE, Novel_TRUE and Novel_FALSE). To account for non-face recognition ability, we added the z-transformed scores of the Cambridge Car Memory Test as a predictor to the second model. Regarding threshold, the two models as well as the results of the linear regression are available in the Supplementary materials. The bayesian linear models were fitted using the brms R package (Bürkner, 2017).

Table 2. Formula specification for the pre-registered linear regression models.

Parameter	Condition	Ability	Formula
Median drift rate	familiarity:accuracy	Face	Median drift rate: 0 + familiarity:accuracy + face_ability + (1|participant)
Median drift rate	familiarity:accuracy	Face + Car	Median drift rate: 0 + familiarity:accuracy + face_ability + car_ability + (1|participant)

familiarity = face familiarity (Learned vs. Novel); accuracy = response accuracy (TRUE vs. FALSE)

2.7.6. Inferential criteria

Using a Bayesian approach in our regression analysis allowed us to evaluate the posterior distributions of the parameter of our predictor of interest (Face Memory score). To evaluate these posterior distributions, we used three complementary criteria: sign, size and precision (Cumming, 2014; Kruschke & Liddell, 2018; McElreath, 2018). To address the sign (or direction) of the effect, we reported if the median value is positive or negative. Second, we evaluated the size or magnitude of the posterior distribution by considering the size of the median value as well as the extent to which lower or upper bounds of the 95% quantile interval approach zero (no relationship). For example, if the 95% quantile interval excluded zero, we would have been reasonably confident in our inference of a non-zero effect. Furthermore, we used the width of the 95% quantile intervals as an indicator of estimation precision. Whereas narrow estimates indicate precise estimates, wider intervals indicate substantial uncertainty about the estimated parameter value, which may warrant a more cautious interpretation.

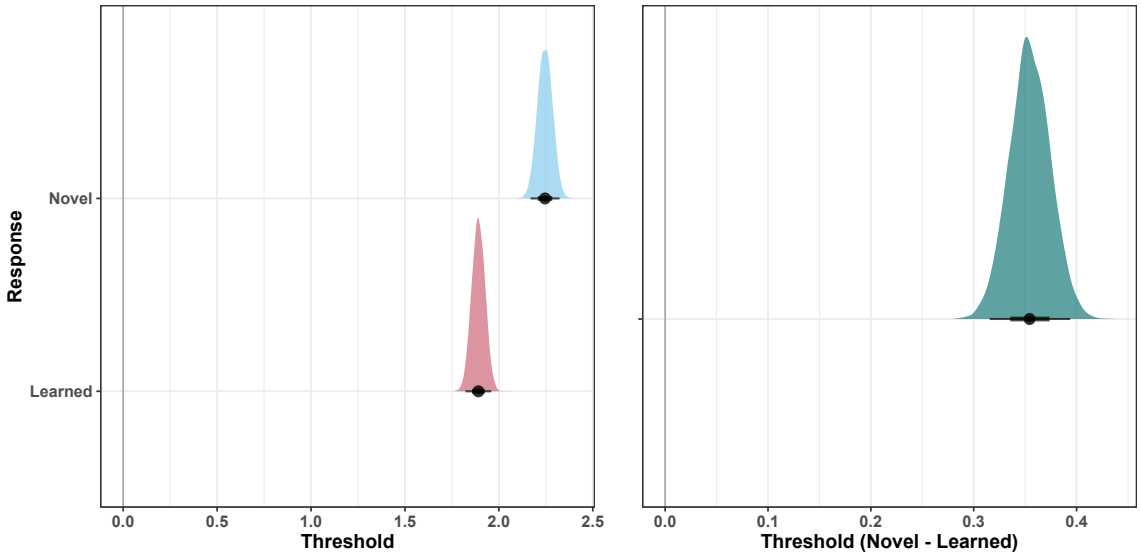
3. Results

We preregistered to assess the relationships between the two recognition abilities (Face and Car) and the LBA parameters (drift rate and threshold). Since our research questions focus on the associations between face recognition ability only and these LBA parameters, we are showing the results of these associations in this Results section. The results of the associations between car recognition ability and LBA parameters can be found in Supplementary Materials.

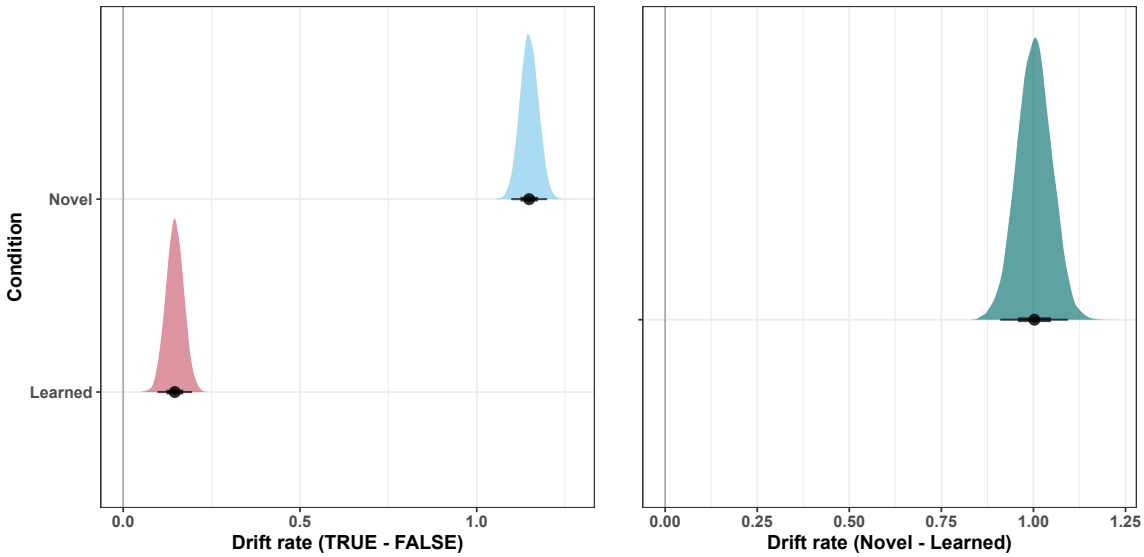
3.1. Estimates of the LBA parameters

In this section, we describe the posterior distributions of the drift rate and threshold parameters for both Learned and Novel faces estimated from the distribution of response times and accuracy collected in the face recognition task. These distributions reflect the group-level effects of the two stimulus condition on drift rate (learned vs novel) and response conditions on threshold (learned vs novel). Contrasts between conditions were plotted to assess the difference between them as well as to establish that the basic experimental effects found in Fournier et al. (2026) was replicated at the group level with larger sample size. Posterior distributions for threshold and drift rate parameters are visualized in Figure 4 and the medians and 95% quantile intervals are summarized in Table 3.

Threshold: Inspection of the posterior distributions shows that there is an effect of familiarity on the threshold parameter. The threshold parameter is higher for the Novel response compared to the Learned response.



(a) Left panel: Threshold by response / Right panel: Threshold (novel - learned) contrast



(b) Left panel: Drift rate (TRUE - FALSE) by stimulus / Right panel: Drift rate (novel - learned) contrast

Figure 4. Threshold and drift rate estimates

Drift rate: Inspection of the posterior distributions shows that there is a familiarity effect in drift rate difference. The quality of the evidence accumulation was greater for Novel faces compared to Learned faces.

Table 3. Posterior median and 95% quantile intervals for threshold and drift rate parameters.

Parameter	Condition	Median	95% QI
Threshold	Novel	2.25	[2.169, 2.323]
Threshold	Learned	1.89	[1.822, 1.96]
Threshold contrast	Novel - Learned	0.35	[0.316, 0.394]
Drift rate	Novel	1.15	[1.098, 1.199]
Drift rate	Learned	0.15	[0.097, 0.195]
Drift rate contrast	Novel - Learned	1	[0.91, 1.094]

QI = quantile interval.

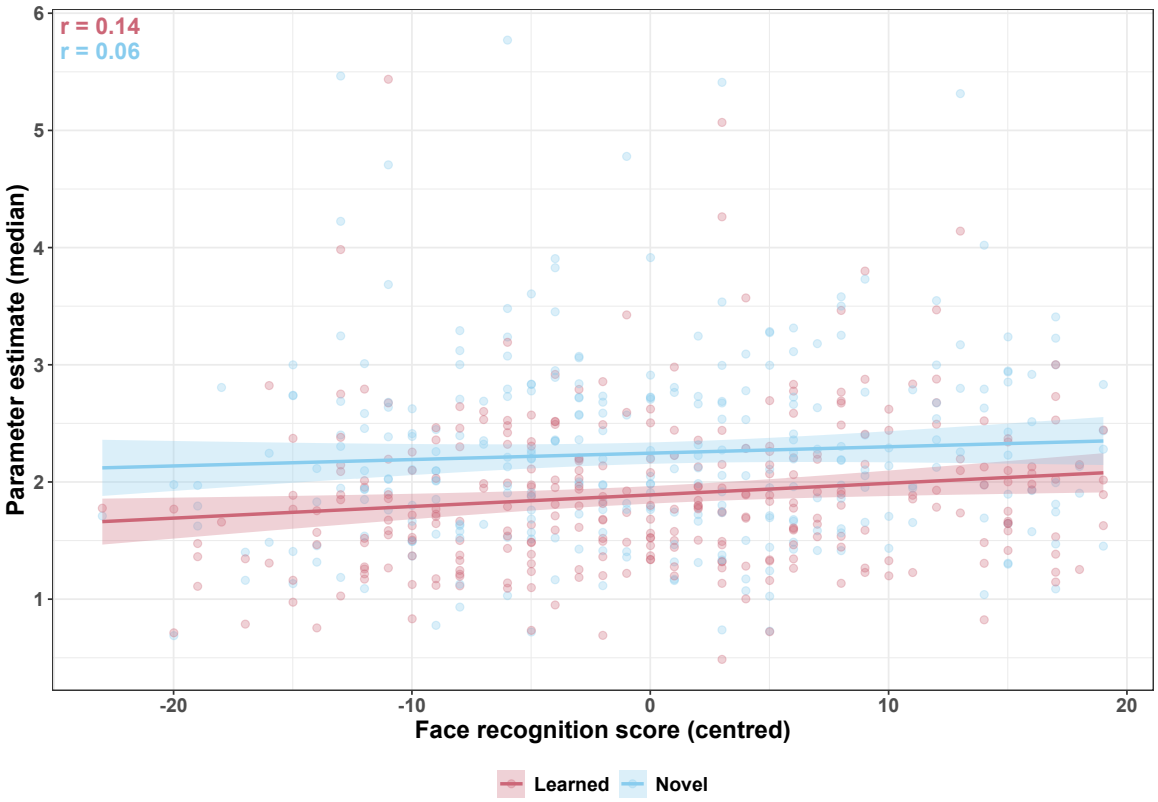
3.2. Classical correlation analysis

To estimate the association between LBA parameters and face recognition ability, we computed the linear relationship between the median of the posterior distribution of each estimated parameters of interest and the face recognition ability scores centred (z-transformed). These relationships can be seen in Figure 5.

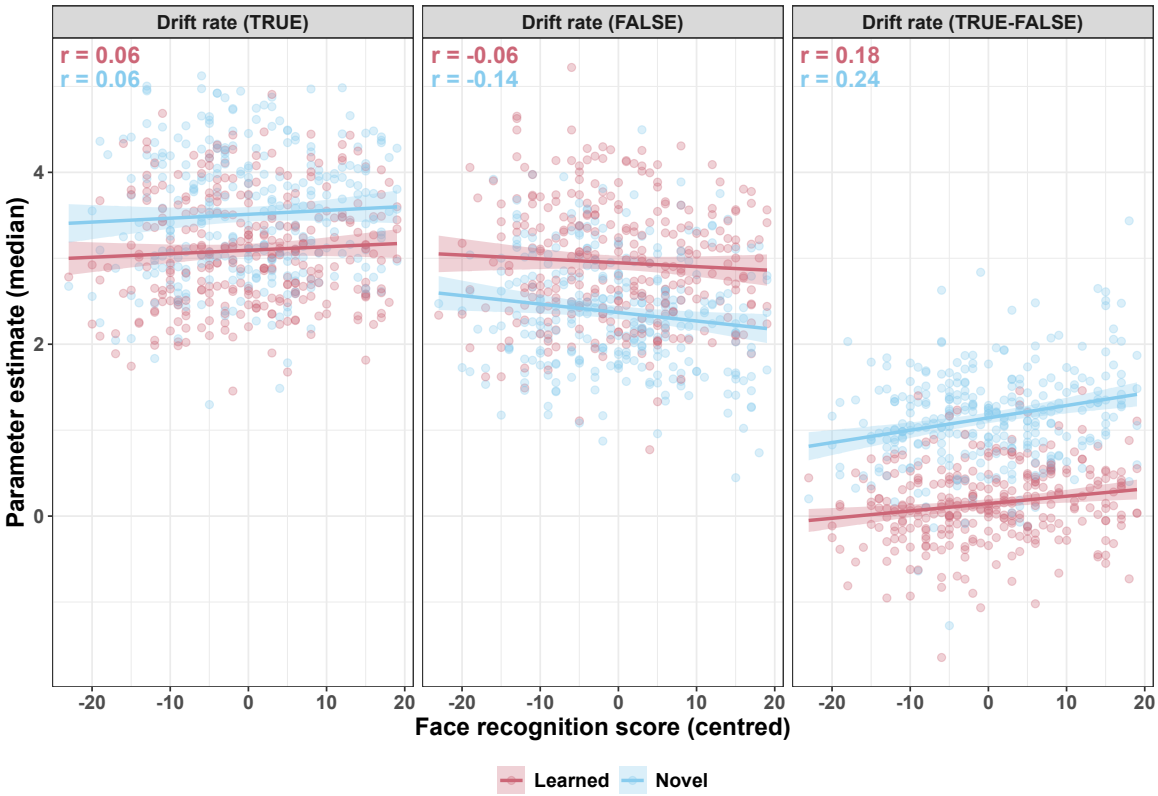
Threshold: Visual inspection of the point estimates correlations shows that when face recognition ability increases, the threshold for learned faces, reflecting the response caution towards the "learned" response, increases. The response caution towards novel faces only seems to increase when car recognition ability increases.

Drift rate: Visual inspection of the point estimates correlations shows that when face recognition ability increases, the drift rate (TRUE-FALSE) for learned and novel faces, reflecting the signal-to-noise ratio provided by learned faces, increases as well.

However, the strength of these relationship remains weak (the highest r value is 0.24).



(a) Median threshold correlations with face recognition ability



(b) Median drift rates correlations with face recognition ability

Figure 5. Point estimates correlations with recognition scores. In (b), drift rate (TRUE) reflects the accumulation of evidence when the response is correct and drift rate (FALSE) the accumulation of evidence when the response is incorrect. Drift rate (TRUE-FALSE) represent the signal-to-noise ratio provided by the stimulus.

3.3. Bayesian plausible value correlational analysis

To estimate the uncertainty linked to the relationships between LBA estimated parameters and face recognition ability, we used a Bayesian plausible value correlational analysis. Like described above (see Methods), instead of using a single point estimate to assess the strength of the relationship between our variables of interest, the posterior distribution of the LBA parameters allows us to use the entire sampling distribution to do it. We calculated the Person's *r* between each sample value from our posterior distribution for each parameter of each subject and their face recognition ability. The posterior distributions of threshold plausible correlation values are shown in Figure 6. Posterior correlations for the different drift rate parameters for both conditions and face recognition ability can be found in Figure 7. For both threshold and drift rate, we evaluated the 95% CI of the posterior distributions. If the 95% CI of the distribution does not include 0, we would be confident in the existence of a relationship between these parameters and the face recognition ability. Table 4 provides a summary of these distributions.

Threshold: There is a positive relationship between response caution for both learned and novel faces and recognition ability. People with a higher face recognition ability have a higher response caution for learned faces.

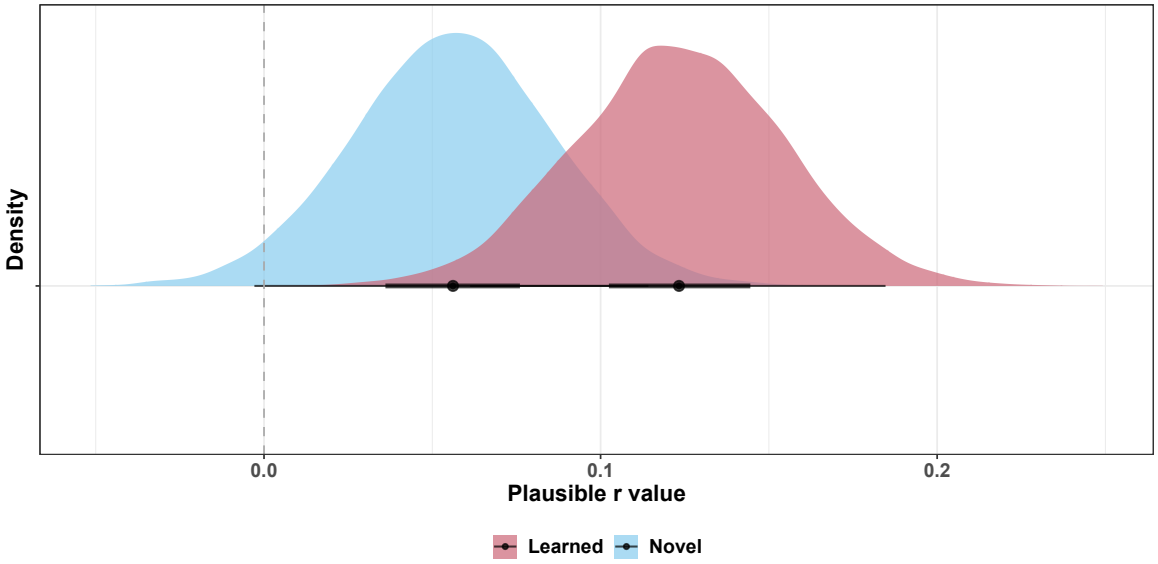
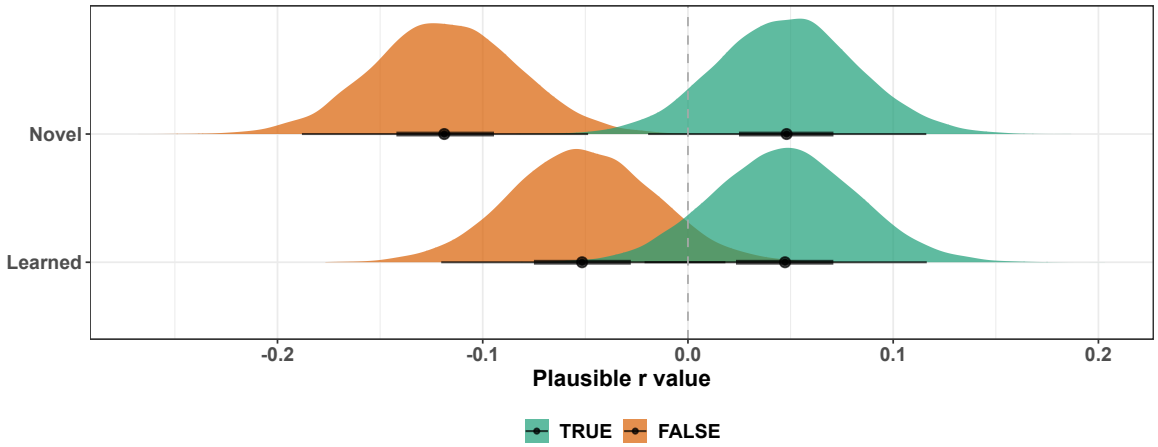
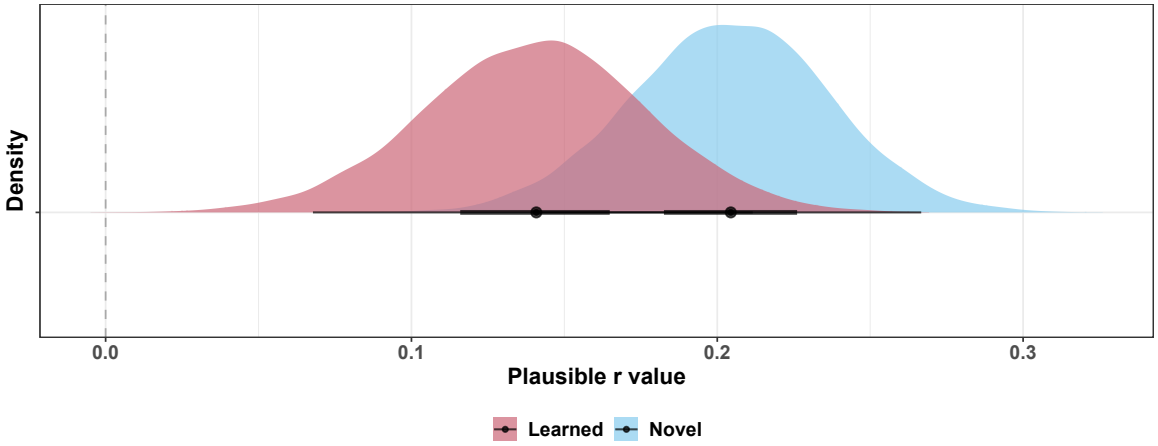


Figure 6. Plausible correlation values distributions between threshold by response and recognition abilities. Points denote the median of the distribution and error bars represent 95% quantile intervals.

Drift rate: Regarding drift rate, face recognition ability is positively associated with the signal-to-noise ratio (drift rate TRUE-FALSE) provided by both novel and learned faces.



(a) Plausible value correlations between drift rate (TRUE) and drift rate (FALSE) and face recognition ability for both stimulus conditions



(b) Plausible value correlations between drift rate (TRUE-FASLE) and face recognition ability for both stimulus conditions

Figure 7. Plausible correlation values distributions between drift rate parameters by stimulus and face recognition ability. Points denote the median of the distribution and error bars represent 95% quantile intervals.

Overall these posterior plausible value correlations show similar observations than the point estimates correlations. However, by taking into account the uncertainty linked to the estimation of the LBA parameters, the confidence in the existence of these associations is increased.

Table 4. Median r and 95% quantile intervals of the plausible value correlations between LBA parameters and face recognition ability scores.

Parameter	Condition	Accuracy	median r	95% QI
Threshold	Learned	-	0.12	[0.06, 0.18]
Threshold	Novel	-	0.06	[−0.00, 0.11]
Drift	Learned	TRUE	0.05	[−0.02, 0.12]
Drift	Learned	FALSE	−0.05	[−0.12, 0.02]
Drift	Learned	TRUE-FALSE	0.14	[0.07, 0.21]
Drift	Novel	TRUE	0.05	[−0.02, 0.12]
Drift	Novel	FALSE	−0.12	[−0.19, −0.05]
Drift	Novel	TRUE-FALSE	0.20	[0.14, 0.27]

Note. Blue = 95% QI entirely above zero; red = 95% QI entirely below zero.

3.4. Linear regression

Following our preregistration, the predictions of the two regression models for drift rate can be seen in Figure 8. From visual inspection, the accumulation of signal coming from the stimulus (drift rate TRUE) seems to increase when face recognition ability increases whereas the accumulation of noise (drift rate FALSE) decreases as a function of face recognition ability for both Novel and Learned faces. This pattern of prediction remains the same when accounting for car recognition ability. Inspection of the posterior predictions in Table 5 shows that some of these predicted intervals overlap with 0, and that caution is needed when extrapolating these results.

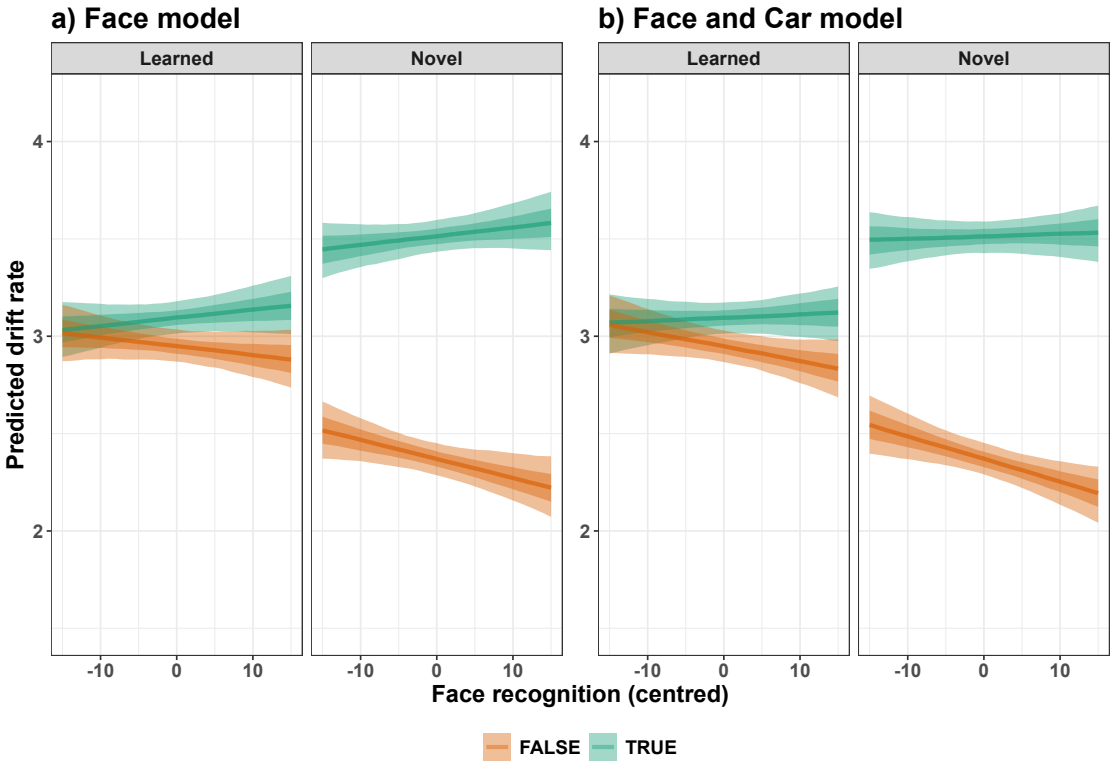


Figure 8. Predicted drift rate values (TRUE and FALSE) from face recognition ability scores for a) Face model and b) Face and Car model

Table 5. Estimates and 95% quantile intervals of the predicted drift rate values slope (TRUE and FALSE) from face recognition ability scores

Parameter	Model	Condition	Accuracy	Estimate	l-95% CI	u-95% CI
Drift slope	Face	Learned	FALSE	−0.0044	−0.0126	0.0039
Drift slope	Face	Novel	FALSE	−0.0097	−0.0179	−0.0014
Drift slope	Face	Learned	TRUE	0.0042	−0.0040	0.0124
Drift slope	Face	Novel	TRUE	0.0047	−0.0035	0.0129
Drift slope	Face + Car	Learned	FALSE	−0.0076	−0.0161	0.0005
Drift slope	Face + Car	Novel	FALSE	−0.0117	−0.0201	−0.0035
Drift slope	Face + Car	Learned	TRUE	0.0017	−0.0068	0.0100
Drift slope	Face + Car	Novel	TRUE	0.0012	−0.0073	0.0093

In Figure 9, the predicted signal-to-noise ratio (drift rate TRUE-FALSE) for both Novel and Learned faces increases as a function of face recognition ability, even when accounting for car recognition ability. As shown in Table 6, these predicted intervals do not overlap with zero. In summary, as people get better at recognizing faces, their ability to accumulate signal and ignoring noisy information in order to recognize Learned faces and to reject Novel faces increases.

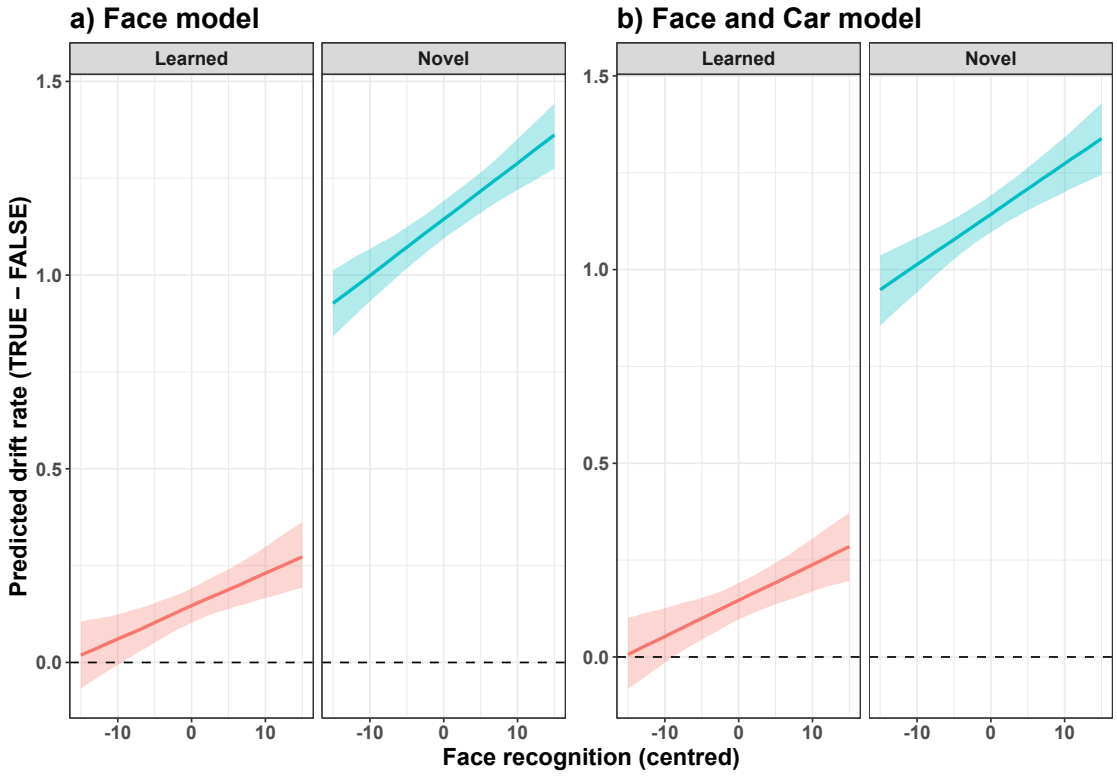


Figure 9. Predicted drift rate values (TRUE-FALSE) from face recognition ability scores for a) Face model and b) Face and Car model

Table 6. Estimates and 95% quantile intervals of the predicted drift rate values slope (TRUE-FALSE) from face recognition ability scores

Parameter	Model	Condition	Estimate	l-95% CI	u-95% CI
Drift slope (TRUE–FALSE)	Face	Learned	0.0086	0.0036	0.0135
Drift slope (TRUE–FALSE)	Face	Novel	0.0144	0.0094	0.0194
Drift slope (TRUE–FALSE)	Face + Car	Learned	0.0092	0.0041	0.0144
Drift slope (TRUE–FALSE)	Face + Car	Novel	0.0129	0.0078	0.0179

4. Discussion

Although accurate recognition of other people is a fundamental ability that guides social behaviour, we currently have a rather limited understanding of the cognitive and computational processes that support such abilities. Using a computational model of decision-making, our results show that higher face recognition ability is associated with multiple latent parameters that subserve face recognition, including the ability to extract meaningful information from a face, as well as the tendency to respond in a more cautious manner. In the following, we situate these findings in the context of the expertise and face recognition literatures, and we comment on the strengths and weaknesses of the approach more generally for future theory development and empirical investigation.

The drift rate results suggest that people who are generally better at recognising faces simultaneously accumulate more signal and less noise from the stimulus, and thus they have a higher drift rate discrimination between true and false accumulators. These findings are in accordance with previous expertise research from a range of domains that showed that expertise effects are consistently due to better information extraction (Dutilh et al., 2009; 2011; Litchfield & Donovan, 2017; Parker et al., 2026; Reinhartz et al., 2023; Thompson et al., 2013; Van Der Gijp et al., 2014). The size of the association (slope) between face recognition ability and drift rate discrimination appeared to be largely invariant to whether the face was visually familiar or entirely novel. Indeed, although the overall drift rate difference was higher for novel than learned faces, which replicates our prior experimental work (Fournier et al., 2026), the association across participants was largely similar. This suggests that, at least in this task configuration using visual familiarity, greater face recognition ability was approximately equally beneficial to accumulating signal and filtering noise for learned and novel faces.

The drift rate results also appeared to be largely similar after we adjusted our regression model for a more general, and non-social measure of perceptual ability (car recognition). This suggests that the relationships we observed between drift rate and face recognition ability cannot easily be explained by a general perceptual ability that generalises to non-social categories. Instead, we suggest that the association is, at least partly, tied to face processing per se, and thus to some degree reflective of a domain-specific process tied the face recognition.

In addition to drift rate discrimination, face recognition ability was also positively associated with response caution. Previous studies have shown that confidence is itself a decision variable used to regulate future behaviour (Fleming et al., 2018; Folke et al., 2016). The positive association between the threshold parameter and face recognition ability could reflect a better estimation of the task difficulty for people with higher face recognition ability, whose response caution is increased to ensure a better performance given the task demands. Whether better recognisers strategically adjust their response caution during the task, or whether this modulation reflects a more automatic process, remains unknown and would require fitting computational models in which the response threshold can vary across trials. Nonetheless, our results suggest that high face recognition ability is not only associated with superior signal extraction during recognition, recognition success is also associated with greater response caution.

More broadly, by combining approaches from experimental, differential and mathematical psychology, we move the research field into a new direction, which we feel has many benefits for understanding the cognitive processes involved in social perception and cognition more generally. Rather than focus on manifest / observed measures (McCaffery et al., 2018; Noyes et al.,

2018), we are able to more directly probe which psychological constructs of the decision-making process best describe different patterns of performance between participants (Alister et al., 2024; Evans & Wagenmakers, 2019). This has knock-on consequences for the development of theories and the specification of hypotheses by being clearer and more specific about the cognitive and computational processes that support face recognition ability. For example, cognitive models of face perception (Bruce & Young, 1986) could incorporate computational mechanisms involved in face recognition and how these mechanisms vary as a function of face recognition ability. Moreover, future research could then test whether these associations between recognition ability and computational mechanisms (signal-to-noise and response caution) extend to the extreme ends of the ability spectrum, by comparing distinct subgroups such as individuals with prosopagnosia and "super-recognisers". Such work would clarify whether the computational mechanisms operate similarly across the entire range of face recognition ability or whether the extremes reflect qualitatively different processes, and would thereby constrain how theories account for individual differences in face recognition.

Although our primary aim was to investigate how face recognition ability predicts computational processes involved in face perception, we conducted exploratory analyses using car recognition ability as a comparison (see Supplementary Materials). Car recognition ability positively predicted drift rate for novel faces but not for learned faces, and unlike face recognition ability, it did not predict the discrimination between TRUE and FALSE drift rates for learned faces. These findings suggest that car and face recognition abilities relate differently to the computational processes underlying face recognition. We speculate that this pattern may reflect differences in information processing: object recognition relies primarily on featural processing, whereas face recognition increasingly integrates configural processing as faces become familiar (Cabeza & Kato, 2000; Madole & Cohen, 1995; Schwaninger et al., 2002; Tversky & Hemenway, 1984). Featural processing alone may support evidence accumulation for novel face recognition while also introducing noise, whereas combining featural and configural processing may enhance meaningful evidence accumulation and reduce detrimental noise. Future research should test whether these distinct associations reflect differences in underlying processing mechanisms.

We have already outlined how our approach has many strengths that add value to our understanding of the face recognition process, but we also want to acknowledge several limitations. First, the number of trials per cell of the design influences the validity and precision of how model parameters are estimated (Alexandrowicz & Gula, 2020; Lerche et al., 2017; Wagenmakers et al., 2007). Our design (90 trials per condition) followed the minimum recommendations to apply EAM (Lüken et al., 2025), and we showed a clear replication of our prior experimental work (Fournier et al., 2026), which is re-assuring. Nonetheless, adding more trials per condition (e.g., 200 per condition; Boag et al. (2025)) would have increased the robustness of our parameter estimates. Second, the associations that we observed were all relatively small (i.e., the highest r value was 0.24). However, the effects, especially for drift rate, were consistent with previous findings on expertise effects, which provides some conceptual confidence that the results can be interpreted inline with prior work (Litchfield & Donovan, 2017; Thompson et al., 2013; Van Der Gijp et al., 2014).

5. Conclusion

Using a computational model of decision-making, we showed that face recognition ability in the general population is associated with multiple latent components of the recognition process. Better recognisers extracted more diagnostic information from faces, reflected in greater drift rate discrimination for both learned and novel faces, in line with expertise effects reported in other domains. This association was not easily explained by a general perceptual ability, which suggests it is at least partly tied to face processing itself. Better recognisers also required more evidence before responding, which may reflect a better calibration to task demands. Together, these findings suggest that individual differences in face recognition ability are reflected in distinct computational mechanisms of the decision process. Identifying these mechanisms provides a more specific basis for theories of face recognition, and for testing whether they operate similarly across the full spectrum of the face recognition ability.

References

Alexandrowicz, R. W., & Gula, B. (2020). Comparing Eight Parameter Estimation Methods for the Ratcliff Diffusion Model Using Free Software. *Frontiers in Psychology*, *11*, 484737. https://doi.org/10.3389/fpsyg.2020.484737

Alister, M., McKay, T., K, Sewell D, K., & Evans, J., N. (2024). Uncovering the Cognitive Mechanisms Underlying the Gaze Cueing Effect. *Quarterly Journal of Experimental Psychology*, *77*(4), 803–827. https://doi.org/10.1177/17470218231181238

Boag, R. J., Innes, R. J., Stevenson, N., Bahg, G., Busemeyer, J. R., Cox, G. E., Donkin, C., Frank, M. J., Hawkins, G. E., Heathcote, A., & al. (2025). An Expert Guide to Planning Experimental Tasks For Evidence-Accumulation Modeling. *Advances in Methods and Practices in Psychological Science*, *8*(2), 25152459251336127. https://doi.org/10.1177/25152459251336127

Bobak, A. K., Hancock, P. J. B., & Bate, S. (2016). Super-recognisers in Action: Evidence from Face-matching and Face Memory Tasks. *Applied Cognitive Psychology*, *30*(1), 81–91. https://doi.org/10.1002/acp.3170

Bowles, D. C., McKone, E., Dawel, A., Duchaine, B., Palermo, R., Schmalzl, L., Rivolta, D., Wilson, C. E., & Yovel, G. (2009). Diagnosing prosopagnosia: Effects of ageing, sex, and participant–stimulus ethnic match on the Cambridge Face Memory Test and Cambridge Face Perception Test. *Cognitive Neuropsychology*, *26*(5), 423–455. https://doi.org/10.1080/02643290903343149

Brown, S. D., & Heathcote, A. (2008). The simplest complete model of choice response time: Linear ballistic accumulation. *Cognitive Psychology*, *57*(3), 153–178. https://doi.org/10.1016/j.cogpsych.2007.12.002

Bruce, V., & Young, A. (1986). Understanding Face Recognition. *British Journal of Psychology*, *77*(3), 305–327. https://doi.org/10.1111/j.2044-8295.1986.tb02199.x

Burton, A. M., White, D., & McNeill, A. (2010). The Glasgow Face Matching Test. *Behavior Research Methods*, *42*(1), 286–291. https://doi.org/10.3758/BRM.42.1.286

Bürkner, P.-C. (2017). *{brms}: An {R} Package for {Bayesian} Multilevel Models Using {Stan}*. *80*. https://doi.org/10.18637/jss.v080.i01

Cabeza, R., & Kato, T. (2000). Features are Also Important: Contributions of Featural and Configural Processing to Face Recognition. *Psychological Science*, *11*(5), 429–433. https://doi.org/10.1111/1467-9280.00283

Cho, S.-J., Wilmer, J., Herzmann, G., McGugin, R. W., Fiset, D., Van Gulick, A. E., Ryan, K. F., & Gauthier, I. (2015). Item response theory analyses of the Cambridge Face Memory Test (CFMT). *Psychological Assessment*, *27*(2), 552–566. https://doi.org/10.1037/pas0000068

Cumming, G. (2014). The New Statistics: Why and How. *Psychological Science*, *25*(1), 7–29. https://doi.org/10.1177/0956797613504966

DeBruine, L. M. (2022). *webmorphR : Reproducible Stimuli*. Zenodo. https://doi.org/10.5281/zenodo.6570965

DeBruine, L., & Jones, B. (2017). *Face Research Lab London Set* (Version 5) [Computer software]. figshare. https://doi.org/10.6084/m9.figshare.5047666.v5

Dennett, H. W., McKone, E., Edwards, M., & Susilo, T. (2012b). Face Aftereffects Predict Individual Differences in Face Recognition Ability. *Psychological Science*, *23*(11), 1279–1287. https://doi.org/10.1177/0956797612446350

Dennett, H. W., McKone, E., Tavashmi, R., Hall, A., Pidcock, M., Edwards, M., & Duchaine, B. (2012a). The Cambridge Car Memory Test: A task matched in format to the Cambridge Face Memory Test, with norms, reliability, sex differences, dissociations from face memory, and expertise effects. *Behavior Research Methods*, *44*(2), 587–605. https://doi.org/10.3758/s13428-011-0160-2

Dobel, C., Bolte, J., Aicher, M., & Schweinberger, S. (2007). Prosopagnosia Without Apparent Cause: Overview and Diagnosis of Six Cases. *Cortex*, *43*(6), 718–733. https://doi.org/10.1016/S0010-9452(08)70501-X

Donkin, C., Averell, L., Brown, S., & Heathcote, A. (2009). Getting more from accuracy and response time data: Methods for fitting the linear ballistic accumulator. *Behavior Research Methods*, *41*(4), 1095–1110. https://doi.org/10.3758/BRM.41.4.1095

Dowsett, A. J., & Burton, A. M. (2015). Unfamiliar face matching: Pairs out-perform individuals and provide a route to training. *British Journal of Psychology*, *106*(3), 433–445. https://doi.org/10.1111/bjop.12103

Duchaine, B., & Nakayama, K. (2006). The Cambridge Face Memory Test: Results for neurologically intact individuals and an investigation of its validity using inverted face stimuli and prosopagnosic participants. *Neuropsychologia*, *44*(4), 576–585. https://doi.org/10.1016/j.neuropsychologia.2005.07.001

Dutilh, G., Krypotos, A.-M., & Wagenmakers, E.-J. (2011). Task-Related Versus Stimulus-Specific Practice. *Experimental Psychology*, *58*(6), 434–442. https://doi.org/10.1027/1618-3169/a000111

Dutilh, G., Vandekerckhove, J., Tuerlinckx, F., & Wagenmakers, E.-J. (2009). A Diffusion Model Decomposition of the Practice Effect. *Psychonomic Bulletin & Review*, *16*(6), 1026–1036. https://doi.org/10.3758/16.6.1026

Evans, N. J., & Wagenmakers, E.-J. (2019). Evidence Accumulation Models: Current Limitations and Future Directions. *Psyarxiv*. https://doi.org/10.31234/osf.io/74df9

Fleming, S. M., Van Der Putten, E. J., & Daw, N. D. (2018). Neural mediators of changes of mind about perceptual decisions. *Nature Neuroscience*, *21*(4), 617–624. https://doi.org/10.1038/s41593-018-0104-6

Folke, T., Jacobsen, C., Fleming, S. M., & De Martino, B. (2016). Explicit representation of confidence informs future value-based decisions. *Nature Human Behaviour*, *1*(1), 2. https://doi.org/10.1038/s41562-016-0002

Fournier, R., Downing, P., & Ramsey, R. (2026). How Does Familiarity Impact the Computational Processes That Support Face Recognition?. *Psyarxiv*. https://doi.org/10.31234/osf.io/854t9_v1

Fysh, M. C., Stacchi, L., & Ramon, M. (2020). Differences between and within individuals, and subprocesses of face cognition: implications for theory, research and personnel selection. *Royal Society Open Science*, *7*(9), 200233. https://doi.org/10.1098/rsos.200233

Gray, K. L. H., Biotti, F., & Cook, R. (2019). Evaluating object recognition ability in developmental prosopagnosia using the Cambridge Car Memory Test. *Cognitive Neuropsychology*, *36*(1–2), 89–96. https://doi.org/10.1080/02643294.2019.1604503

Haines, N., Kvam, P., Irving, L., Smith, C., Beauchaine, T., Pitt, M., Ahn, W.-Y., & Turner, B. (2020). *Theoretically Informed Generative Models Can Advance the Psychological and Brain Sciences: Lessons from the Reliability Paradox*. https://osf.io/xr7y3_v1

Kruschke, J. K., & Liddell, T. M. (2018). Bayesian data analysis for newcomers. *Psychonomic Bulletin & Review*, *25*(1), 155–177. https://doi.org/10.3758/s13423-017-1272-1

Lerche, V., Voss, A., & Nagler, M. (2017). How many trials are required for parameter estimation in diffusion modeling? A comparison of different optimization criteria. *Behavior Research Methods*, *49*(2), 513–537. https://doi.org/10.3758/s13428-016-0740-2

Lewandowsky, S., & Oberauer, K. (2018). Computational Modeling in Cognition and Cognitive Neuroscience. In J. T. Wixted (Ed.), *Stevens' Handbook of Experimental Psychology and Cognitive Neuroscience* (1st ed., pp. 1–35). Wiley. https://doi.org/10.1002/9781119170174.epcn501

Litchfield, D., & Donovan, T. (2017). The flash-preview moving window paradigm: Unpacking visual expertise one glimpse at a time. *Frontline Learning Research*, *5*(3), 66–80. https://doi.org/10.14786/flr.v5i3.269

Lubinski, D. (2000). Scientific and Social Significance of Assessing Individual Differences: "Sinking Shafts at a Few Critical Points". *Annual Review of Psychology*, *51*(1), 405–444. https://doi.org/10.1146/annurev.psych.51.1.405

Ly, A., Boehm, U., Heathcote, A., Turner, B. M., Forstmann, B., Marsman, M., & Matzke, D. (2017). A Flexible and Efficient Hierarchical Bayesian Approach to the Exploration of Individual Differences in Cognitive-model-based Neuroscience. In A. A. Moustafa (Ed.), *Computational Models of Brain and Behavior* (1st ed., pp. 467–479). Wiley. https://doi.org/10.1002/9781119159193.ch34

Lüken, M., Heathcote, A., Haaf, J. M., & Matzke, D. (2025). Parameter identifiability in evidence-accumulation models: The effect of error rates on the diffusion decision model and the linear ballistic accumulator. *Psychonomic Bulletin & Review*, *32*(3), 1411–1424. https://doi.org/10.3758/s13423-024-02621-1

Madole, K. L., & Cohen, L. B. (1995). The role of object parts in infants' attention to form-function correlations. *Developmental Psychology*, *31*(4), 637–648. https://doi.org/10.1037/0012-1649.31.4.637

Marsh, T., & Boag, S. (2013). Evolutionary and differential psychology: conceptual conflicts and the path to integration. *Frontiers in Psychology*, *4*. https://doi.org/10.3389/fpsyg.2013.00655

Matzke, D., Ly, A., Selker, R., Weeda, W. D., Scheibehenne, B., Lee, M. D., & Wagenmakers, E.-J. (2017). Bayesian Inference for Correlations in the Presence of Measurement Error and Estimation Uncertainty. *Collabra: Psychology*, *3*(1), 25. https://doi.org/10.1525/collabra.78

McCaffery, J. M., Robertson, D. J., Young, A. W., & Burton, A. M. (2018). Individual differences in face identity processing. *Cognitive Research: Principles and Implications*, *3*(1), 21. https://doi.org/10.1186/s41235-018-0112-9

McConachie, H. R. (1976). Developmental Prosopagnosia. A Single Case Report. *Cortex*, *12*(1), 76–82. https://doi.org/10.1016/S0010-9452(76)80033-0

McElreath, R. (2018). *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. Chapman, Hall/CRC. https://doi.org/10.1201/9781315372495

Murray, E., & Bate, S. (2020). Diagnosing developmental prosopagnosia: Repeat assessment using the Cambridge Face Memory Test. *Royal Society Open Science*, *7*(9), 200884. https://doi.org/10.1098/rsos.200884

Myers, A., & Hansen, C. H. (2006). *Experimental Psychology* (6th ed.). Thomson Wadsworth.

Myers, C. E., Interian, A., & Moustafa, A. A. (2022). A Practical Introduction to Using the Drift Diffusion Model of Decision-Making in Cognitive Psychology, Neuroscience, and Health Sciences. *Frontiers in Psychology*, *13*. https://doi.org/10.3389/fpsyg.2022.1039172

Noyes, E., Hill, M. Q., & O'Toole, A. J. (2018). Face recognition ability does not predict person identification performance: using individual data in the interpretation of group results. *Cognitive Research: Principles and Implications*, *3*(1), 23. https://doi.org/10.1186/s41235-018-0117-4

Parker, S., Cross, E. S., & Ramsey, R. (2026). Evidence Accumulation Modelling Offers New Insights Into the Cognitive Mechanisms That Underlie Linguistic and Action-Based Training. *Collabra: Psychology*, *12*(1), 160157. https://doi.org/10.1525/collabra.160157

Ratcliff, R., Smith, P. L., Brown, S. D., & McKoon, G. (2016). Diffusion Decision Model: Current Issues and History. *Trends in Cognitive Sciences*, *20*(4), 260–281. https://doi.org/10.1016/j.tics.2016.01.007

Reinhartz, A., Strobach, T., Jacobsen, T., & Bastian, C. C. von. (2023). Mechanisms of Training-Related Change in Processing Speed: A Drift-Diffusion Model Approach. *Journal of Cognition*, *6*(1), 46. https://doi.org/10.5334/joc.310

Rhodes, G., Jeffery, L., Taylor, L., Hayward, W. G., & Ewing, L. (2014). Individual Differences in Adaptive Coding of Face Identity Are Linked to Individual Differences in Face Recognition Ability. *Journal of Experimental Psychology: Human Perception and Performance*, *40*(3), 897–903. https://doi.org/10.1037/a0035939

Richler, J. J., Tomarken, A. J., Sunday, M. A., Vickery, T. J., Ryan, K. F., Floyd, R. J., Sheinberg, D., Wong, A. C. N., & Gauthier, I. (2019a). Individual differences in object recognition. *Psychological Review*, *126*(2), 226–251. https://doi.org/10.1037/rev0000129

Richler, J. J., Tomarken, A. J., Sunday, M. A., Vickery, T. J., Ryan, K. F., Floyd, R. J., Sheinberg, D., Wong, A. C.-N., & Gauthier, I. (2019b). Individual Differences in Object Recognition. *Psychological Review*, *126*(2), 226–251. https://doi.org/10.1037/rev0000129

Russell, R., Duchaine, B., & Nakayama, K. (2009). Super-recognizers: People with extraordinary face recognition ability. *Psychonomic Bulletin & Review*, *16*(2), 252–257. https://doi.org/10.3758/PBR.16.2.252

Schwaninger, A., Lobmaier, J. S., & Collishaw, S. M. (2002). Role of Featural and Configural Information in Familiar and Unfamiliar Face Recognition. In H. H. Bülthoff, C. Wallraven, S.-W. Lee,

& T. A. Poggio (Eds.), *Biologically Motivated Computer Vision* (Vol. 2525, pp. 643–650). Springer Berlin Heidelberg. https://doi.org/10.1007/3-540-36181-2_64

Shakeshaft, N. G., & Plomin, R. (2015). Genetic specificity of face recognition. *Proceedings of the National Academy of Sciences*, *112*(41), 12887–12892. https://doi.org/10.1073/pnas.1421881112

Simons, D., Shoda, Y., & Lindsay, D. (2017). Constraints on Generality (COG): A Proposed Addition to All Empirical Papers. *Perspectives on Psychological Science*, *12*, 174569161770863. https://doi.org/10.1177/1745691617708630

Stevenson, N., Donzallaz, M., Heathcote, A., & Miletić, S. (2026). *EMC2: Bayesian Hierarchical Analysis of Cognitive Models of Choice*. https://doi.org/10.32614/CRAN.package.EMC2

Thompson, M. B., Tangen, J. M., & McCarthy, D. J. (2013). Expertise in Fingerprint Identification. *Journal of Forensic Sciences*, *58*(6), 1519–1530. https://doi.org/10.1111/1556-4029.12203

Tversky, B., & Hemenway, K. (1984). Objects, parts, and categories. *Journal of Experimental Psychology: General*, *113*(2), 169–193. https://doi.org/10.1037/0096-3445.113.2.169

Van Der Gijp, A., Van Der Schaaf, M. F., Van Der Schaaf, I. C., Huige, J. C. B. M., Ravesloot, C. J., Van Schaik, J. P. J., & Ten Cate, T. J. (2014). Interpretation of radiological images: towards a framework of knowledge and skills. *Advances in Health Sciences Education*, *19*(4), 565–580. https://doi.org/10.1007/s10459-013-9488-y

Verhallen, R. J., Bosten, J. M., Goodbourn, P. T., Lawrance-Owen, A. J., Bargary, G., & Mollon, J. (2017). General and specific factors in the processing of faces. *Vision Research*, *141*, 217–227. https://doi.org/10.1016/j.visres.2016.12.014

Wagenmakers, E.-J., Van Der Maas, H. L. J., & Grasman, R. P. P. P. (2007). An EZ-diffusion model for response time and accuracy. *Psychonomic Bulletin & Review*, *14*(1), 3–22. https://doi.org/10.3758/BF03194023

Wang, R., Li, J., Fang, H., Tian, M., & Liu, J. (2012). Individual Differences in Holistic Processing Predict Face Recognition Ability. *Psychological Science*, *23*(2), 169–177. https://doi.org/10.1177/0956797611420575

White, C. N., & Kitchen, K. N. (2022). On the Need to Improve the Way Individual Differences in Cognitive Function Are Measured With Reaction Time Tasks. *Current Directions in Psychological Science*, *31*(3), 223–230. https://doi.org/10.1177/09637214221077060

White, C. N., Curl, R. A., & Sloane, J. F. (2016). Using Decision Models to Enhance Investigations of Individual Differences in Cognitive Neuroscience. *Frontiers in Psychology*, *7*. https://doi.org/10.3389/fpsyg.2016.00081

White, D., Kemp, R. I., Jenkins, R., Matheson, M., & Burton, A. M. (2014). Passport Officers' Errors in Face Matching. *Plos ONE*, *9*(8), e103510. https://doi.org/10.1371/journal.pone.0103510

Wilmer, J. B., Germine, L., Chabris, C. F., Chatterjee, G., Gerbasi, M., & Nakayama, K. (2012). Capturing specific abilities as a window into human individuality: The example of face recognition. *Cognitive Neuropsychology*, *29*(5–6), 360–392. https://doi.org/10.1080/02643294.2012.753433

Wilmer, J. B., Germine, L., Chabris, C. F., Chatterjee, G., Williams, M., Loken, E., Nakayama, K., & Duchaine, B. (2010). Human face recognition ability is specific and highly heritable. *Proceedings of the National Academy of Sciences*, *107*(11), 5238–5241. https://doi.org/10.1073/pnas.0913053107

Yarkoni, T. (2022). The generalizability crisis. *Behavioral and Brain Sciences*, *45*, e1. https://doi.org/10.1017/S0140525X20001685

Yovel, G., Wilmer, J. B., & Duchaine, B. (2014). What Can Individual Differences Reveal about Face Processing?. *Frontiers in Human Neuroscience*, *8*. https://doi.org/10.3389/fnhum.2014.00562